

AutoEDA Pro — Statistical & Profiling Report

Dataset: 10a21ac3-9881-4101-a677-f0865233b181 • Autonomous Machine Learning Telemetry

1. Dataset Overview

Total Rows	0	Total Columns	0
Missing Cells (%)	0.00%	Target Column	N/A

2. Executive Summary Narrative

Executive EDA & Dataset Summary Report

Target Directory: `C:\Users\Anish Kumar Verma\PycharmProjects\AutoEDA\sandbox\_run\10a21ac3-9881-4101-a677-f0865233b181`

Processed Files: `agent\_plan\_log.json`, `agent\_state.json`, `current\_df.csv`, `metadata\_profile.json`, `metrics.json`

Excluded Files: `generated\_analysis.py` (Script excluded from summary)

1. Dataset Overview

- **Dataset Identifier:** `StudentsPerformance.csv`

- **Dimensions:** `1000` rows x `9` columns

- **Target Variable:** `total\_score`

- **Data Quality:** No missing values detected in raw profile.

1.5 Full Column Statistics

Column	Type	Missing %	Unique %	Mean	Median	Std	Skew	Kurtosis
gender	str	0.0%	0.2%	N/A	N/A	N/A	N/A	N/A
race/ethnicity	str	0.0%	0.5%	N/A	N/A	N/A	N/A	N/A
parental level of education	str	0.0%	0.6%	N/A	N/A	N/A	N/A	N/A
lunch	str	0.0%	0.2%	N/A	N/A	N/A	N/A	N/A
test preparation course	str	0.0%	0.2%	N/A	N/A	N/A	N/A	N/A
math score	int64	0.0%	8.1%	66.09	66.0	15.16	-0.28	0.27
reading score	int64	0.0%	7.2%	69.17	70.0	14.6	-0.26	-0.07
writing score	int64	0.0%	7.7%	68.05	69.0	15.2	-0.29	-0.03

2. Data Imputation & Preprocessing

- **status:** Imputation completed

3. Outlier Analysis (IQR Method)

No numeric outlier statistics reported.

4. Derived Domain Attributes & Composite Metrics

- **total_score**: Formula: $\text{math score} + \text{reading score} + \text{writing score}$ | Purpose: Aggregated academic performance across all subjects.

5. Statistical Hypothesis Testing & Key Predictors

All predictors below were tested against `total_score` and found statistically significant ($p < 0.05$), ranked by effect size.

| Feature | Test Type | Effect Size | Label | P-Value | Why It Matters |

|---|---|---|---|---|---

| `reading score` | Pearson Correlation | 0.9703 | Strong correlation | 0.0000e+00 | High literacy skills are almost always linked to higher overall academic performance across all subjects. |

| `writing score` | Pearson Correlation | 0.9657 | Strong correlation | 0.0000e+00 | Strong writing ability is a primary indicator of a student's total success in the curriculum. |

| `math score` | Pearson Correlation | 0.9187 | Strong correlation | 0.0000e+00 | Numerical proficiency is a core component that consistently tracks with a student's cumulative grade. |

| `lunch` | ANOVA | 0.2901 | Large effect | 7.7368e-21 | Access to nutritional support programs is associated with significant differences in overall student achievement levels. |

| `test preparation course` | ANOVA | 0.2567 | Large effect | 1.6338e-16 | Completing specialized preparatory programs relates to higher total scores compared to students who do not participate. |

| `parental level of education` | ANOVA | 0.2265 | Large effect | 4.3810e-10 | The educational background of parents is a key factor tied to the academic outcomes of their children. |

| `race/ethnicity` | ANOVA | 0.1878 | Large effect | 3.2259e-07 | Diverse demographic backgrounds show distinct patterns in total academic performance across the student population. |

| `gender` | ANOVA | 0.1309 | Medium effect | 3.3120e-05 | Differences in overall scores are observed between male and female students across the dataset. |

6. Redundancy & Multicollinearity Analysis

Numeric-Numeric High Correlation Pairs ($|r| \geq 0.85$):

| Feature 1 | Feature 2 | Correlation (r) | Interpretation |

|---|---|---|---

| `math score` | `total_score` | 0.9187 | Strong correlation |

| `reading score` | `writing score` | 0.9546 | Strong correlation |

| `reading score` | `total_score` | 0.9703 | Strong correlation |

| `writing score` | `total_score` | 0.9657 | Strong correlation |

7. Generated Visual Artifacts

No PNG/SVG image assets found in directory.

8. Categorical Associations (Cramer's V)

| Feature 1 | Feature 2 | Cramer's V |

|---|---|---|

| `gender` | `race/ethnicity` | 0.0709 |

| `gender` | `parental level of education` | 0.0 |

| `gender` | `lunch` | 0.0 |

| `gender` | `test preparation course` | 0.0 |

| `race/ethnicity` | `parental level of education` | 0.0487 |

| `race/ethnicity` | `lunch` | 0.0 |

| `race/ethnicity` | `test preparation course` | 0.0385 |

| `parental level of education` | `lunch` | 0.0 |

| `parental level of education` | `test preparation course` | 0.0674 |

| `lunch` | `test preparation course` | 0.0 |

9. Predictive Modeling Strategy Blueprint

- **Target Definition:** total_score

- **Problem Type:** Regression

Recommended Algorithms

- Regularized Linear Regression (Ridge / Lasso)

- Random Forest Regressor

- Gradient Boosting Regressor

- Support Vector Regressor (SVR)

Feature Selection Strategy

- Exclude high-cardinality ID or text name columns

- Rank features using cross-validated permutation importance and mutual information

- Remove collinear features exceeding correlation threshold > 0.85

Validation Strategy

- K-Fold Cross-Validation (5 folds)

- Evaluate MAE, RMSE, R-Squared, and Residual Error distribution

Overfitting Risk Mitigation

- Apply regularization penalties (L1/L2)

- Limit tree depth and enforce minimum samples per leaf

- Perform hyperparameter tuning strictly within cross-validation folds

- **Executive Summary:** Target: 'total_score' (Regression). Model recommendations and validation strategy tailored for 1000 rows x 9 columns.

Report generated automatically by `summary\_generator.py`

4. Statistical Hypothesis Testing Results

Feature	Test Conducted	Statistic	P-Value	Significant
reading score	Pearson Correlation	0.9703	0.0000e+00	YES ($\alpha=0.05$)
writing score	Pearson Correlation	0.9657	0.0000e+00	YES ($\alpha=0.05$)
math score	Pearson Correlation	0.9187	0.0000e+00	YES ($\alpha=0.05$)
lunch	ANOVA	0.2901	7.7368e-21	YES ($\alpha=0.05$)
test preparation course	ANOVA	0.2567	1.6338e-16	YES ($\alpha=0.05$)
parental level of education	ANOVA	0.2265	4.3810e-10	YES ($\alpha=0.05$)
race/ethnicity	ANOVA	0.1878	3.2259e-07	YES ($\alpha=0.05$)
gender	ANOVA	0.1309	3.3120e-05	YES ($\alpha=0.05$)

5. Predictive Modeling Strategy Blueprint

Target: 'total_score' (Regression). Model recommendations and validation strategy tailored for 1000 rows x 9 columns.

Task Type	N/A
Recommended Models	N/A
Validation Strategy	5-Fold Stratified K-Fold